Two counterexamples for stationary MMPP arrival times

Research note for review — 2 October 2026

Abstract

Conjecture 3.9 of Asanjarani and Nazarathy, published in the ANZIAM Journal in 2022, asserts that the time-stationary first-arrival time of an irreducible Markov-modulated Poisson process stochastically dominates its event-stationary first-arrival time. We exhibit an irreducible 16-state directed-cycle model with rational transition and arrival rates for which the asserted survival inequality is reversed at time 12. A self-contained analytic argument proves a violation exceeding 5.5×10^{-6}, and a separate exact rational matrix-series certificate verifies the sign. The same rational model also answers negatively He's separate question whether the time-stationary first-arrival time must have squared coefficient of variation at least one. This leaves intact the event-stationary squared-coefficient-of-variation result established by He. The mathematical verification below does not assert priority; independent literature and human review remain necessary before any publication claim.

1 The conjecture and the model

We use row-vector conventions for probability distributions. An irreducible finite-state Markov-modulated Poisson process (MMPP) consists of a continuous-time Markov chain J with generator Q, and a Poisson arrival process whose conditional instantaneous rate is d_{J_t}. Write

$$D = \operatorname{diag}(d_0, \dots, d_{15}), \qquad C = Q - D, \qquad \pi Q = 0, \quad \pi\mathbf{1} = 1.$$

The mean arrival rate and the event-stationary state distribution are

$$\overline{\lambda} = \pi D\mathbf{1}, \qquad \alpha = \frac{\pi D}{\overline{\lambda}}. \tag{1}$$

For an MMPP, arrivals do not change the state of J, so α is also the state distribution immediately after an arrival in the stationary arrival process. If T^β is the time to the next arrival when J_0 has distribution β, its survival function is

$$F_\beta(t) = \mathbb{P}(T^\beta > t) = \beta e^{tC}\mathbf{1} = \mathbb{E}_\beta \exp\left(-\int_0^t d_{J_s}\, ds\right). \tag{2}$$

Conjecture 3.9 of [1] states $T^\pi \geq_{\mathrm{st}} T^\alpha$ for every irreducible MMPP, equivalently

$$F_\pi(t) \geq F_\alpha(t) \qquad (t \geq 0). \tag{3}$$

Equation (1) and the survival inequality are equations (2.1) and (2.9), respectively, in that paper. No reversibility of Q is assumed in its conjecture.

Index the states modulo 16. Set

$$Q_{i,i} = -1, \qquad Q_{i,i+1} = 1, \qquad Q_{i,j} = 0 \quad (j \notin \{i, i+1\}). \tag{4}$$

This generator is irreducible, and its stationary distribution is $\pi_i = 1/16$. Define the integer vector

$$\begin{aligned}(s_0, \dots, s_{15}) = (&1000, 924, 707, 383, 0, -383, -707, -924,\\ &-1000, -924, -707, -383, 0, 383, 707, 924)\end{aligned} \tag{5}$$

and choose the arrival rates

$$d_i = \frac{10000 + s_i}{10^6} = \lambda + \varepsilon g_i, \qquad \lambda = \frac{1}{100}, \quad \varepsilon = \frac{1}{1000}, \quad g_i = \frac{s_i}{1000}. \tag{6}$$

Since $\sum_i s_i = 0$, we have $\overline{\lambda} = \lambda$ and

$$\alpha_i = \frac{10000 + s_i}{160000}.$$

Every arrival rate lies in $[9/1000, 11/1000]$, so the model has strictly positive rates. In particular, both π and α are probability distributions with strictly positive entries.

Theorem 1. *For the irreducible MMPP (4)–(6),*

$$F_\pi(12) - F_\alpha(12) < -5.544 \times 10^{-6}. \tag{7}$$

Thus this model is a counterexample to (3).

2 Survival, hazard, and perturbation identities

We first record identities valid for any finite-state stationary MMPP with positive arrival rates. Since $\pi C = -\pi D$, differentiating (2) gives

$$F'_\pi(t) = \pi C e^{tC}\mathbf{1} = -\pi D e^{tC}\mathbf{1} = -\overline{\lambda} F_\alpha(t). \tag{8}$$

This uses the stationary identity $\pi Q = 0$, not reversibility. The hazard rate from the time-stationary distribution therefore satisfies

$$h_\pi(t) := -\frac{F'_\pi(t)}{F_\pi(t)} = \overline{\lambda}\frac{F_\alpha(t)}{F_\pi(t)}. \tag{9}$$

Consequently, the asserted order (3) is equivalent to $h_\pi(t) \le \overline{\lambda}$ for all $t \ge 0$.

For the explicit model, take $J_0 \sim \pi$ and put

$$A_t = \int_0^t g(J_s)\, ds, \qquad H(t) = \mathbb{E}_\pi e^{-\varepsilon A_t}. \tag{10}$$

Then $F_\pi(t) = e^{-\lambda t} H(t)$, and (8) yields the exact identity

$$F_\pi(t) - F_\alpha(t) = \frac{e^{-\lambda t}}{\lambda} H'(t). \tag{11}$$

The mean of $g(J_t)$ is zero and $|g(J_t)| \le 1$.

Lemma 1. *For every $t \geq 0$,*

$$H'(t) = \varepsilon^2 I_g(t) + R(t), \qquad I_g(t) = \int_0^t \mathbb{E}_\pi[g(J_0)g(J_s)]\, ds, \tag{12}$$

where

$$|R(t)| \leq \frac{\varepsilon^3 t^2}{2} e^{\varepsilon t}. \tag{13}$$

Proof. For each fixed t, the chain has probability zero of jumping at exactly t. Since $|A_{t+h} - A_t| \leq |h|$, dominated differentiation in (10) is justified and gives

$$H'(t) = -\varepsilon \mathbb{E}_\pi[g(J_t)e^{-\varepsilon A_t}].$$

The elementary Taylor formula

$$e^{-x} = 1 - x + x^2 \int_0^1 (1-u)e^{-ux}\, du$$

implies $|e^{-x} - 1 + x| \leq x^2 e^{|x|}/2$ for real x. Using $\mathbb{E}_\pi g(J_t) = 0$ and $|A_t| \leq t$ therefore gives

$$H'(t) = \varepsilon^2 \mathbb{E}_\pi[g(J_t)A_t] + R(t)$$

with the bound (13). By stationarity and Fubini,

$$\mathbb{E}_\pi[g(J_t)A_t] = \int_0^t \mathbb{E}_\pi[g(J_s)g(J_t)]\, ds = \int_0^t \mathbb{E}_\pi[g(J_0)g(J_{t-s})]\, ds = I_g(t).$$

The equality uses scalar multiplication and a forward time shift; no time reversal of the chain is needed. □

3 A self-contained analytic certificate

Let $\theta = \pi/8$ and $f_i = \cos(i\theta)$. For the directed-cycle chain (4), one can write $J_t = J_0 + N_t \pmod{16}$, where N_t is a rate-one Poisson process independent of J_0. Averaging over $J_0 \sim \pi$ gives

$$\frac{1}{16}\sum_{i=0}^{15} \cos(i\theta)\cos((i+n)\theta) = \frac{1}{2}\cos(n\theta).$$

The Poisson generating function consequently yields

$$\mathbb{E}_\pi[f(J_0)f(J_t)] = \frac{1}{2}e^{-at}\cos(bt), \qquad a = 1 - \cos\theta, \quad b = \sin\theta. \tag{14}$$

Here $a^2 + b^2 = 2a$. Integrating (14) gives

$$I_f(t) := \int_0^t \mathbb{E}_\pi[f(J_0)f(J_s)]\, ds = \frac{a + e^{-at}[-a\cos(bt) + b\sin(bt)]}{4a}. \tag{15}$$

All terminating decimals below denote exact rational numbers. The standard bounds $3.141 < \pi < 3.142$ and the half-angle formulas give

$$.075 < a < .077, \qquad .382 < b < .384. \tag{16}$$

For clarity, these radical estimates need no numerical approximation: $1.414^2 < 2 < 1.415^2$, and

$$\cos^2\theta = \frac{2+\sqrt{2}}{4}, \qquad \sin^2\theta = \frac{2-\sqrt{2}}{4}.$$

Comparison with $.923^2, .925^2$ proves the bounds on a; comparison with $.382^2, .384^2$ proves the bounds on b.

Set $\delta = 3\pi/2 - 12b$. Equation (16) gives

$$.1035 < \delta < .129 < .13.$$

Thus $\cos(12b) = -\sin\delta \geq -.13$ and

$$\sin(12b) = -\cos\delta \leq -\left(1 - \frac{\delta^2}{2}\right) < -.99.$$

In particular,

$$X := -a\cos(12b) + b\sin(12b) < .077 \cdot .13 - .382 \cdot .99 = -.36817. \tag{17}$$

Since $12a < .924 < 1$ and $e < 3$, we have $e^{-12a} > 1/3$. The fact that X is negative matters when multiplying inequalities:

$$a + e^{-12a}X < .077 - \frac{.36817}{3} = -\frac{.13717}{3}.$$

Since $4a < .308$, equation (15) now gives

$$I_f(12) < -\frac{.13717}{.924} < -.148. \tag{18}$$

The last comparison is exact: $.148 \cdot .924 = .136752 < .13717$. The estimate $e < 3$ follows, for example, from $\sum_{n\geq 2} 1/n! < \sum_{n\geq 2} 2^{-(n-1)} = 1$.

We next transfer this sign to the rational vector g.

$$\|g - f\|_\infty < \frac{1}{2000}. \tag{19}$$

To verify every entry, it suffices by symmetry to consider $1, \cos\theta, \cos(\pi/4), \sin\theta, 0$. The exact square comparisons $1.4142^2 < 2 < 1.4143^2$ give

$$.9235 < \cos\theta < .9245, \qquad .7065 < \cos(\pi/4) < .7075, \qquad .3825 < \sin\theta < .3835.$$

These prove (19) for the entries $.924, .707, .383$ and hence for all 16 entries. For any two times u, v, since $|f|, |g| \leq 1$,

$$|g(J_u)g(J_v) - f(J_u)f(J_v)| \leq |g(J_u) - f(J_u)|\,|g(J_v)| + |f(J_u)|\,|g(J_v) - f(J_v)| < \frac{1}{1000}.$$

It follows that $|I_g(12) - I_f(12)| < .012$. In conjunction with (18), this proves

$$I_g(12) < -.136. \tag{20}$$

Finally, the remainder in Lemma 1 satisfies

$$\frac{|R(12)|}{\varepsilon^2} \leq .072e^{.012} \leq \frac{.072}{1 - .012} = \frac{18}{247} < .073.$$

Here $e^x \le (1-x)^{-1}$ for $0 \le x < 1$ follows by comparing their power series term by term. Therefore

$$\frac{H'(12)}{\varepsilon^2} < -.136 + .073 = -.063. \tag{21}$$

Using $e^{-.12} > 1 - .12 = .88$ in (11) gives

$$F_\pi(12) - F_\alpha(12) < -.063 \cdot 10^{-6} \cdot \frac{.88}{.01} = -5.544 \times 10^{-6}.$$

This completes the proof of Theorem 1 without relying on floating-point evaluation of a matrix exponential.

4 An independent exact rational matrix certificate

The same sign can be verified directly from the rational model. Set $w = \pi - \alpha$, $r = 1011/1000$, and

$$B = 12(C + rI).$$

The only nonzero entries of B are

$$B_{i,i} = 12\left(\frac{11}{1000} - d_i\right) \ge 0, \qquad B_{i,i+1} = 12.$$

Hence B is entrywise nonnegative and

$$L := \|B\|_\infty = \frac{1503}{125} = 12.024, \qquad \|w\|_1 = \frac{1257}{20000}.$$

Define the exact rational numbers

$$S_{80} = \sum_{k=0}^{80} \frac{wB^k\mathbf{1}}{k!}, \qquad E_{80} = \|w\|_1 \frac{L^{81}}{81!}\frac{1}{1 - L/82}. \tag{22}$$

The matrix norm inequality gives $|wB^k\mathbf{1}| \le \|w\|_1 L^k$. For $k \ge 81$, consecutive scalar terms $L^k/k!$ have ratio at most $L/82 < 1$. Summing this geometric majorant proves

$$|we^B\mathbf{1} - S_{80}| \le E_{80}. \tag{23}$$

No cancellation assumption is used in the tail bound.

The companion file `mmpp_conjecture39_check.py` evaluates (22) using Python's standard-library rational type `fractions.Fraction`. It verifies the three exact inequalities

$$E_{80} < 10^{-31}, \qquad -4 < S_{80} - E_{80}, \qquad S_{80} + E_{80} < -\frac{39}{10}. \tag{24}$$

Its sparse recurrence can also be reproduced independently. Start with the row vector $u^{(0)} = w$ and, for $k = 1, \dots, 80$, set

$$u_j^{(k)} = \frac{u_j^{(k-1)} B_{j,j} + 12u_{j-1}^{(k-1)}}{k} \qquad (j \pmod{16}).$$

Then $u^{(k)} = wB^k/k!$ and $S_{80} = \sum_{k=0}^{80}\sum_j u_j^{(k)}$. Every operation is rational. For orientation, the exact rational quantities have decimal displays

$$S_{80} \approx -3.9292649153377824801, \qquad E_{80} \approx 3.8718957312181850 \times 10^{-35}.$$

These decimal displays are not used to certify (24). Combining (23) and (24) gives

$$-4 < we^B\mathbf{1} < -\frac{39}{10}, \qquad F_\pi(12) - F_\alpha(12) = e^{-12r}we^B\mathbf{1} < 0.$$

For a rational negative bound from this certificate alone, $12r < 13$ and $e < 3$ imply

$$F_\pi(12) - F_\alpha(12) < -\frac{39}{10 \cdot 3^{13}} = -\frac{13}{5314410} < -2.4 \times 10^{-6}.$$

An ordinary numerical evaluation gives

$$F_\pi(12) - F_\alpha(12) \approx -2.1156832900682 \times 10^{-5}.$$

The analytic and exact rational certificates suffice independently of this approximation.

5 Time-stationary squared coefficient of variation

Section 4.3 of [2] also asks whether the time-stationary first-arrival time always has squared coefficient of variation at least one. The same rational model provides a negative answer.

Proposition 1. *For the MMPP (4)–(6),*

$$0 < \mathrm{CV}(T^\pi)^2 < \frac{199999}{200000} < 1.$$

Proof. Write $R = (D - Q)^{-1}$. Since all arrival rates are at least $9/1000$, (2) implies that $e^{tC}\mathbf{1} \le e^{-9t/1000}\mathbf{1}$. Thus the matrix integral $\int_0^\infty e^{tC}\,dt$ converges, equals R, and has nonnegative entries. Integration of the survival function and the semigroup identity give the usual phase-type moment formulas

$$\mu := \mathbb{E}_\pi T = \pi R\mathbf{1}, \qquad m_2 := \mathbb{E}_\pi T^2 = 2\pi R^2\mathbf{1}.$$

For completeness, the second formula follows from $R^2 = \int_0^\infty \int_0^\infty e^{(s+t)C}\,ds\,dt = \int_0^\infty te^{tC}\,dt$. Let $u = R\mathbf{1}$ and $z = Ru$. Their entries are rational and can be computed using the following explicit cyclic recurrence. For any rational right-hand side b, the system $(D - Q)x = b$ is

$$a_i x_i - x_{i+1} = b_i, \qquad a_i = 1 + d_i > 1, \qquad i \pmod{16}.$$

Put $P = \prod_{i=0}^{15} a_i$. Since $P > 1$, the unique solution starts at

$$x_0 = \frac{\displaystyle\sum_{j=0}^{15} b_j \prod_{k=j+1}^{15} a_k}{P - 1}, \qquad x_{i+1} = a_i x_i - b_i.$$

Empty products equal one. Iterating the recurrence gives $x_{16} = Px_0 - \sum_j b_j \prod_{k=j+1}^{15} a_k = x_0$, verifying the cyclic boundary condition. Apply it first to $b = \mathbf{1}$ to obtain u, and then to $b = u$ to obtain z.

The companion exact rational verifier evaluates $\mu = \pi u$ and $m_2 = 2\pi z$ and checks

$$\mu^2 < m_2 < \frac{399999}{200000}\mu^2.$$

Dividing by $\mu^2 > 0$ proves the assertion because $\mathrm{CV}(T^\pi)^2 = m_2/\mu^2 - 1$. In addition to the cyclic recurrence, independent exact rational Gaussian elimination gives the same values. Their approximate decimal displays are

$$\mu \approx 100.0027991122752187, \qquad m_2 \approx 20001.06092113849143, \qquad \mathrm{CV}(T^\pi)^2 \approx .9999941263846573.$$

The strict rational comparisons, rather than these decimal displays, are the certificate. □

6 Scope and interpretation

The sign mechanism is the negative integrated autocovariance of a nonreversible cycle. For a small modulation of the rate, this quantity is the leading term in $H'(t)$, and therefore determines which side of the mean rate the time-stationary hazard occupies. The calculation above controls all higher-order terms for an explicit rational model.

Theorem 1 disproves the universal survival-order statement of Conjecture 3.9 in [1], and Proposition 1 answers negatively the time-stationary variability question in Section 4.3 of [2]. Neither result refutes the distinct assertion that the event-stationary interarrival time has squared coefficient of variation at least one. That assertion is established in [2]; Section 4.3 there treats the stronger stochastic-order question as a further research question. A survival inequality at every time is stronger than an inequality between these moments. The directed-cycle example also makes no claim about reversible modulating chains.

This is a specific published conjecture from 2019/2022, rather than a famous decades-old open problem. The results and their exact certificates are presented for review. No claim that this is the first such counterexample is made; an independent check of subsequent literature and of the argument is appropriate before submission or announcement.

References

[1] A. Asanjarani and Y. Nazarathy, *Stationary Markovian Arrival Processes: Results and Open Problems*, The ANZIAM Journal **64** (2022), 54–68. doi:10.1017/S1446181122000013. Preprint: arXiv:1905.01736 (2019).

[2] Q.-M. He, *Bounds on the mean and squared coefficient of variation of phase-type distributions*, Journal of Applied Probability **58** (2021), 880–889. doi:10.1017/jpr.2021.18.